

Q.1 Explain the role of vision in industrial robots.

Ans: Vision plays a crucial role in industrial robots by enabling them to see, recognize and interact with their environment. Using cameras and image processing algorithms, robots can identify object shapes, sizes, positions, and orientations. Vision helps robots perform tasks such as inspection, assembly, quality control, sorting, and guidance. It improves accuracy, flexibility, and automation in manufacturing process.

Q.2 Define digital image representation. Explain grey scale and RGB images.

Ans: A digital image is represented as a two-dimensional matrix of pixels, where each pixel holds an intensity value.

- Grayscale Image: Each pixel represents brightness, only, usually ranging from 0-Black to 255 (white). It is stored as a single matrix.

- RGB Image: Each pixel has three components: Red, Green, Blue, each ranging from 0 to 255. It is

Stored as three matrices combined to form a color image.

Q.3 What is spatial resolution? How does it impact robotic perception?

Ans: Spatial resolution refers to the number of pixels used to represent an image, typically expressed as width $\times$ height.

- Higher resolution - more detail & accuracy
- Lower resolution - loss of fine details.

In robotic perception, higher spatial resolution allows robots to better detect edges, small objects, and fine features, improving recognition and decision-making accuracy.

Q.4 Explain the working principle of a camera used in robotic systems.

Ans: A camera captures light reflected from objects through a lens & focuses it onto an image sensor (CCD or CMOS). The sensor converts light into electrical signals, which are digitized into pixel values. These pixel values form a digital image that is processed

by the robot's vision system to extract useful information such as object position, shape & movement.

Q.5 Convert a 3×3 grayscale image into matrix form and explain intensity values.

Ans: A grayscale image can be represented in matrix form where each pixel element corresponds to the intensity value of a pixel. For a 3×3 grayscale image, the image consists of 9 pixels arranged in three rows and three columns.

eg:
$$\begin{bmatrix} 50 & 100 & 150 \\ 80 & 120 & 200 \\ 30 & 90 & 255 \end{bmatrix}$$

Each value in the matrix represents the brightness of a pixel. Lower intensity values indicate darker shades, while higher values represent brighter shades. This matrix representation allows the robot's vision system to process & analyze the image numerically for tasks such as texture detection and object recognition.

Q.6 Analyze how perspective projection helps a robot estimate object distance.

Ans. Perspective projection models how a 3D object is projected onto a 2D image plane. Objects closer to the camera appear larger, while distant objects appear smaller. By analyzing object size, position, and camera parameters, a robot can estimate the distance of objects. This helps robots perform tasks such as navigation, obstacle avoidance and object manipulation.

Q.7 Study a real-time vision-based robot application (line follower, pick-and-place, or face recognition) and explain the vision ~~pipeline~~ pipeline.

Ans. Example: Vision Pipeline of a Line-Follower Robot.

- Image Acquisition:

A camera mounted on the robot continuously captures images of the path or line on the ground.



- Preprocessing:

The captured images are converted to grayscale and filtered to remove noise, improving image clarity for further analysis.

- Image Segmentation:

Thresholding techniques are applied to separate the line from the background based on pixel intensity differences.

- Feature Extraction:

The system identifies important features such as the position & direction of the line within the image.

- Decision Making & Control:

Based on the extracted features, control commands are generated to adjust the robot's motors, enabling smooth and accurate line following in real time.

Q.8 Evaluate challenges in implementing vision-based systems in real-time robots & propose solutions using modern AI-based vision techniques.



Ans: Challenges:

- High computational requirements
- Sensitivity to lighting changes
- Real-time processing constraints.
- Occlusion & noisy data.

AI-based solution:

- Use of deep learning models (CNNs) for robust feature extraction.
- Adaptive vision algorithms to handle lighting variations
- Edge computing & GPUs for faster processing
- Sensor fusion ~~combining~~ combining vision with LiDAR or depth sensor.

Modern AI-based vision techniques significantly improve accuracy, speed, & reliability in real-time robotics systems.